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ORIGINAL ARTICLE

Parametric study and metallurgical investigation of AISI 308LSi wire arc additive manufacturing single beads

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Abstract
This study presents a parametric and metallurgical investigation of single bead AISI 308LSi stainless steel fabricated by wire arc additive manufacturing. While process optimization has traditionally focused primarily on bead geometry, the combined influence of key process parameters on metallurgical characteristics and corrosion behavior remains insufficiently understood. Therefore, the effects of welding current, travel speed, and shielding gas flow rate on bead geometry, microstructure, microhardness, and corrosion resistance were investigated. Grey relational analysis was used to identify the optimal parameter combination. Silicon improved bead penetration, while travel speed strongly affected bead geometry, and current primarily controlled penetration depth. The optimal parameters were 130 A, 0.48 m/min, and 20 L/min. X-ray diffraction confirmed a predominantly γ austenitic microstructure. Electron Backscatter Diffraction indicated that grain structure evolution was governed by process parameters through their effects on thermal gradients and solidification rates. Lower current combined with higher travel speed promoted grain refinement and increased the fraction of Σ3 twin boundaries, whereas higher current favored columnar grain growth and increased the heterogeneity of the columnar and equiaxed grain distribution. The existence of equiaxed grains at the top of the bead highlighted the influence of cooling due to the shielding gas. Microhardness decreased with increasing current and varied with grain structure, whereas a refined grain-boundary network with a higher Σ3 twin boundary fraction at an intermediate heat input of 278 J/mm enhanced passive film stability and, consequently, corrosion resistance. Optimized process parameters enabled microstructural control while enhancing microhardness and corrosion resistance.

Keywords WAAM · Heat input · Bead geometry · Microstructure · GRA · EBSD

1 Introduction

Additive manufacturing (AM) has emerged as a key technology in the evolution of Industry 4.0, transforming the design and fabrication of complex components across various sectors [1], It offers several advantages, including rapid prototyping, high dimensional accuracy, efficient material utilization, and the elimination of conventional tooling. According to ISO/ASTM 52900:2021 [2], AM is defined as a process of joining materials to produce parts from 3D model data, typically layer by layer [3]. These attributes make AM particularly attractive for aerospace, biomedical,

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energy, and automotive industries. Metal AM comprises several techniques, mainly differentiated by the type of feedstock. Powder-based processes such as Laser Powder Bed Fusion (LPBF), are widely used for producing intricate geometries with high precision and excellent surface finish. In contrast, wire-based methods like Wire Arc Additive Manufacturing (WAAM) offer a more cost-effective solution for fabricating large-scale components. WAAM relies on arc-welding techniques such as Gas Metal Arc Welding (GMAW), Gas Tungsten Arc Welding (GTAW), or Cold Metal Transfer (CMT) to melt metallic wire, allowing for high deposition rates (up to 10 kg/h depending on the material) and minimal material waste. These features make WAAM suitable for structural applications in aerospace, marine, and energy sectors, as well as for part repair and remanufacturing [4–6]. Recent advances in metal additive manufacturing have extended beyond conventional parameter optimisation to include the use of reinforcement phases, multi-material architectures, and numerical approaches for controlling thermal conditions and material response. Carbon-based reinforcements, including carbon nanotubes and graphene, have been incorporated into additively manufactured metallic matrices to modify microstructural evolution and improve mechanical and tribological behaviour [7, 8]. Similarly, additively manufactured bimetallic structures have been investigated to understand interfacial characteristics and the mechanical response of dissimilar material combinations [9, 10]. In parallel, numerical studies have examined the influence of heat input and post-deposition heat treatment on thermal history, phase evolution, residual stress, and distortion in additively manufactured metallic alloys [11, 12]. Collectively, these studies demonstrate that material performance in AM depends on the coupled effects of composition, thermal conditions, process parameters, and post-processing.

Based on the definition provided in [2], the successful fabrication of large additively manufactured components relies fundamentally on the quality of each deposited layer. Consequently, achieving a stable and well-controlled bead geometry is essential to ensure the integrity of the final build. However, controlling bead geometry remains a complex task. In the (CMT) process, several parameters can be adjusted, including current (I), travel speed (TS), and shielding gas flow rate (GFR). Among the various stainless steel grades employed in AM applications, AISI 308L has attracted considerable attention due to its combination of good corrosion resistance, adequate mechanical performance [13]. Numerous studies have therefore investigated the optimal combination of these parameters to achieve desirable bead geometries. Le et al. [14] investigated the influence of I, voltage (U), and TS on the geometry of single beads produced by WAAM using the GMAW process. Their objective was to optimize process parameters for the deposition of AISI 308L stainless steel walls. A Box–Behnken [15] design of experiments was employed to analyze the effect of the variables. The results indicated that voltage contributes approximately 70% to bead width (BW), while deposition speed accounts for nearly 50% of the variation in bead height. Similarly, Abioye et al. [16] investigated the effect of Wire Feed Rate (WFR) and TS on bead geometry and mechanical properties of WAAM-fabricated 308L stainless steel using a factorial experimental design. Their results showed that increasing WFR leads to a larger bead width due to the higher volume of molten material supplied to the melt pool. Bead height increases up to a certain limit before decreasing as a result of melt pool instability. In contrast, increasing TS decreases both the deposited material per unit length and the heat input (HI), resulting in smaller bead dimensions. Additionally, hardness was found to be strongly influenced by both WFR and TS, with the highest hardness obtained at low WFR due to the precipitation of Cr-rich phases. Karamimoghadam et al. [17] investigated the influence of three key process parameters: wire feed rate, current, and arc voltage on the surface roughness, microhardness, and microstructure of AISI 308L stainless steel fabricated using WAAM-CMT. Their results showed that columnar austenite grows along the heat extraction direction. Grain refinement occurs near the surface due to higher cooling rates, resulting in finer grains and increased austenite content. Increasing the printing current improves fusion and molten metal spreading, producing a smoother surface. In contrast, lower current leads to rougher surfaces and incomplete fusion. Electron Backscatter Diffraction (EBSD) analysis revealed a preferential grain growth along the ⟨111⟩ crystallographic direction. In addition, grain size varies across the bead and decreases from the substrate toward the deposited region due to thermal gradients and mixing at the fusion boundary. The microhardness distribution follows this microstructural evolution. Higher hardness is observed at the center of the fusion zone, where finer grains and δ-ferrite enrichment are present. Lower hardness is found at the bead edges and top, where coarser grains are observed. Morvayova et al. [18] evaluated the influence of arc power and deposition speed on temperature evolution and bead geometry during WAAM-CMT fabrication of AISI 308L stainless steel. A design of experiments based on a 2^3 full factorial design was employed. The results showed that increasing arc power leads to a quasi-linear increase in bead width. It also reduces the wetting angle due to higher heat input (HI) and improved melt spreading. Response surface analysis further indicated that arc power is the dominant parameter influencing bead width and wetting angle, whereas deposition speed has no significant effect on bead geometry. Previous studies have established that the geometry of WAAM

deposited AISI 308L beads is mainly governed by process variables such as current or arc power, travel speed, voltage, and wire-feed rate. They have also shown that the thermal conditions during deposition influence phase distribution, grain morphology, surface condition, and hardness. However, most published studies have focused on conventional AISI 308L, whereas the process–structure–property relationships of AISI 308LSi under CMT-WAAM conditions remain comparatively underexplored. This distinction is relevant because the higher Si content of AISI 308LSi may affect melt-pool behaviour, wetting, and corrosion-related performance. In addition, existing optimisation approaches frequently prioritise individual geometric responses, such as bead width or bead height, rather than considering geometry, microstructural characteristics, hardness, and corrosion resistance concurrently.

Accordingly, the current status of the field shows that WAAM parameter effects on 308L bead geometry and selected mechanical or microstructural characteristics are relatively well documented; however, an integrated assessment of CMT-WAAM AISI 308LSi that links parameter selection to bead geometry, microstructure, microhardness, and corrosion resistance is still limited. The present work addresses this gap through a Taguchi-based experimental design in which current, travel speed, and shielding gas flow rate were selected from literature-supported processing ranges and evaluated at three levels. Grey Relational Analysis was then used to optimise the multiple bead-geometry responses while minimising heat input. The resulting condition was validated using X-ray diffraction, optical microscopy, scanning electron microscopy, EBSD, microhardness testing, and corrosion measurements. By establishing parameter–geometry–microstructure–property relationships for CMT-WAAM AISI 308LSi single beads, this work provides a basis for the design and optimisation of subsequent multi-layer WAAM builds.

The previous studies provide valuable insights into the process structure relationships governing WAAM fabricated stainless steels. From the literature, it can be inferred that the deposition of AISI 308L single beads using GMAW or CMT based WAAM has been widely investigated. These studies consistently demonstrate that bead geometry is primarily governed by key processing parameters such as current (or arc power), TS and wire feed rate. Furthermore, the resulting microstructure and hardness are strongly influenced by the thermal conditions during deposition, particularly heat input and cooling rate, which in turn govern the balance between δ-ferrite and austenite phases. Despite these advances, several gaps remain in the literature. Most studies focus exclusively on conventional AISI 308L, whereas AISI 308LSi has received limited attention despite its higher silicon content, which may influence melt pool dynamics and wetting behavior and decreases the intensity of intergranular corrosion [19], this alloy is also widely used in several industrial sectors, including oil and gas, pharmaceutical equipment, and the automotive industry [20]. Furthermore, the current state of art shows that WAAM parameter effects on 308L bead geometry and selected mechanical or microstructural characteristics are relatively well documented. However, a detailed analysis of CMT-WAAM AISI 308LSi that links parameter selection to bead geometry, microstructure, microhardness, and corrosion resistance is still limited. The present work addresses this gap through a Taguchi-based experimental design in which current, travel speed, and shielding gas flow rate were selected from literature.A Grey Relational Analysis (GRA) approach [21] was employed to simultaneously optimize bead geometry while minimizing the heat input. The optimization results are subsequently validated through physical characterization using X-Ray Diffraction (XRD), Optical and Scanning Electron Microscopy (OM/SEM), EBSD, and corrosion testing. The optimized parameter set proposed in this work provides a reliable reference for future WAAM applications and process development including AISI 308LSi.

2 Materials design and experimental methodology

WAAM experiments were performed using a semi-automatic additive manufacturing system equipped with a Fronius CMT 500i power source coupled to a three-axis CNC positioning system showed in Fig. 1a. The feedstock material was AISI 308LSi austenitic stainless steel wire (ESAB) with a diameter of 1.2 mm. The chemical composition of the wire is presented in Table 1. Single beads were deposited on an AISI 316L substrate, as shown in Fig. 1b, with a deposited bead length of 10 cm. The selection of the processing parameters was based on a comprehensive literature review summarized in Table 2, which compiles previous WAAM studies on AISI 308L stainless steel using CMT and GMAW processes. Based on these studies, lower and upper bounds were identified for the main parameters governing single-bead formation. The parameter ranges adopted in the present study were: (I): 130–160 A, (TS): 0.28–0.48 m/min, and (GFR): 15–20 L/min, using an Ar + 2.5% CO_2 shielding gas mixture. The choice of current, TS, and GFR as the main process variables was motivated by their direct influence on arc stability, heat input, and bead geometry within the operational window of the CMT process.

An intermediate value within each parameter range was selected, resulting in three levels for each factor, as presented in Table 3. A full factorial design would require 3^3 (i.e., 27) parameter combinations. Therefore, to reduce the

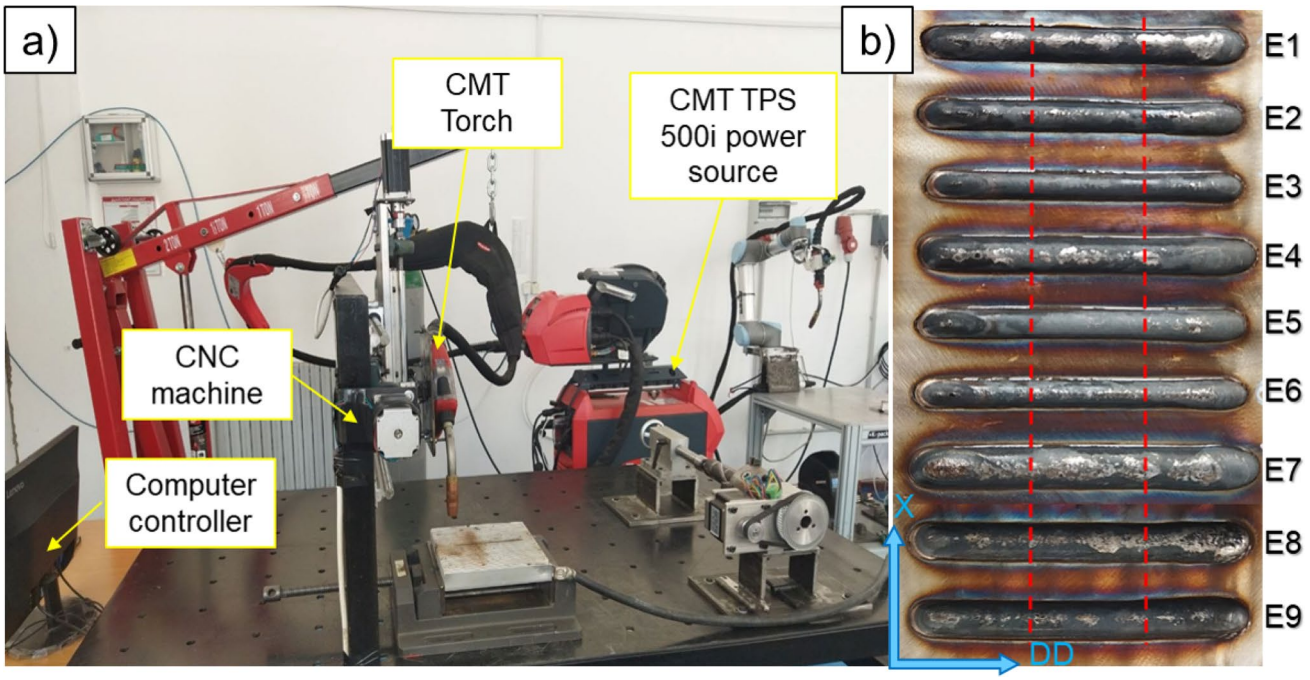


Fig. 1 **a** Fronius CMT TPS 500i power source coupled to a computer-controlled three-axis (X, Y, and Z) CNC positioning system. **b** deposited single beads with a length of 10 cm. The red dashed lines indicate the locations where the samples were sectioned for analysis

Table 1 Nominal chemical composition (wt %) of the commercial AISI 308LSi stainless steel filler wire according to the manufacturer's technical data sheet

Chemical component	C	Mn	Si	Ni	Cr	Mo	Cu	Ferrite FN
%	0.01	1.8	0.9	10.5	19.9	0.15	0.1	9

Table 2 Summary of WAAM Process Parameters for 308LSi Reported in the Literature

Reference	Process	TS (m/min)	I (A)	v (V)	Shield gas	GFR(L/min)
Anand et al. [20]	CMT	0.368	155	20	(Ar+2.5%CO_2)	13
Chen et al. [22]	CMT	0.48	155–160	20.2	(Ar+3%CO_2)	20
Koli et al. [23]	CMT	0.48	140	16	Ar 99.99%	16
Nagasai et al. [24]	CMT	0.28	132	14.6	Ar 99.99%	15
Li et al. [25]	CMT	0.36–0.42	135–150	19–21	(Ar+2.5%CO_2)	/
Haldar et al. [26]	CMT	0.368	133	19	Ar+2.5% CO_2	13

Table 3 Selected levels of travel speed, current, and shielding gas flow rate for WAAM Trials

Level	TS (m/min)	I (A)	GFR (L/min)
Level 1	0.28	130	15
Level 2	0.38	145	17.5
Level 3	0.48	160	20

number of experimental runs while maintaining a balanced parametric evaluation, a Taguchi L9 orthogonal array design of experiments (DoE) was adopted [23]. This reduced the number of experiments to nine experiments (Ei, i = 1–9), as shown in Table 4. To analyze the bead geometry, the deposited samples were sectioned at two locations along the bead length, positioned 3 cm from each end of the bead. The specimens were prepared by mechanical polishing using abrasive papers up to 2400 grit, followed by fine polishing with alumina suspension to obtain a mirror-like surface. The samples were then chemically etched using a solution of HNO_3, HCl, and H_2O (1:2:3). Samples for microstructural analysis were prepared by the same methodology. The macrographs of the cross-sections were obtained using Nikon Eclipse LV 100ND light optical microscope. The geometric parameters measured, namely bead width (BW), bead height (BH), contact angle (CA), and melt-in depth (D), are presented in Fig. 2a and were considered the main responses for the parametric study. Measurements were performed using ImageJ software. At least two measurements were taken for each parameter, and the reported value corresponds to their average. The relatively uniform bead morphology across the different experimental conditions justified the selection of only two measurement positions for the analysis.

Table 4 Summary of process parameters, bead geometry and GRG rankings for WAAM trials

Experience N°	I (A)	TS (m/min)	GFR (L/min)	Heat input (J/mm)	BH (mm)	BW (mm)	Contact angle °	Melt-in depth (mm)	GRG	Rank
E1	130	0.28	15	307	3.76	9.13	104.3	1.35	0.55	7
E2	130	0.38	17.5	226	3.25	8.42	118.8	1.39	0.62	3
E3	130	0.48	20	179	2.74	8.09	131.1	1.23	0.80	1
E4	145	0.28	17.5	377	3.80	9.97	115.2	1.62	0.53	9
E5	145	0.38	20	278	3.38	8.52	121.5	1.57	0.59	5
E6	145	0.48	15	220	2.99	7.81	124.2	1.49	0.70	2
E7	160	0.28	17.5	447	3.93	11.14	123.2	2.31	0.57	6
E8	160	0.38	20	329	3.59	9.77	121.2	1.85	0.54	8
E9	160	0.48	15	260	2.96	9.35	129.4	1.90	0.60	4

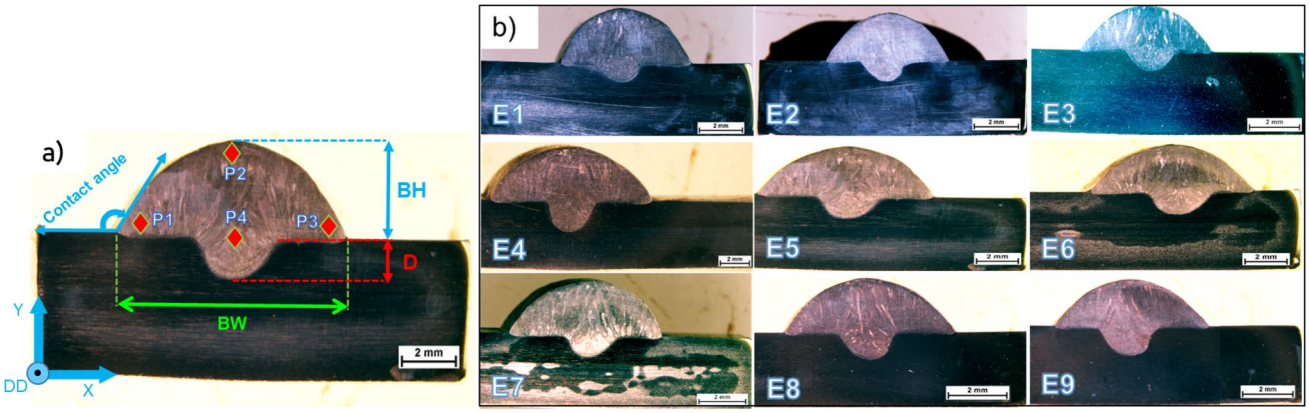


Fig. 2 **a** Cross-section of a deposited bead illustrating the measured geometrical parameters, namely bead height (BH), bead width (BW), contact angle (CA), and melt-in depth (D). Positions P1 to P4 correspond to the locations used for microhardness measuremen

The response values associated with each set of process parameters, as well as the HI calculated using Eq. (1) [27], are summarized in Table 4.

$$Hi = \frac{V.I.\eta}{TS}, \tag{1}$$

where V is the average arc voltage (V), I is the average current (A), η is the process thermal efficiency (0.8 for CMT), and TS is the travel speed (mm/s).

According to the literature [28, 29], the selection of fabrication parameters should be guided by the desired output responses (i.e., high (BH) and low (BW)). To achieve this, Grey Relational Analysis (GRA) [30] was applied to optimize the process parameters based on multiple responses. Experimental results were converted into mean values and evaluated using appropriate criteria (larger-the-better or smaller-the-better), depending on the response type. X-Ray diffraction (XRD) analysis was performed using a Bruker D2 PHASER diffractometer with Co radiation (λ = 1.78897 Å) over a 2θ range of 40°–120°. Electron backscatter diffraction (EBSD) analysis was carried out using a Zeiss Sigma SEM equipped with an Oxford Instruments detector. Measurements were performed using an accelerating voltage of 20 kV with a step size of 4 µm. EBSD maps were analyzed using MTEX 6.0.0. For the nine specimens, data were extracted from a region located sufficiently far from the substrate to avoid the influence of substrate grains on the quantitative analysis. The grain area fraction was determined based on an aspect ratio criterion. Grains with L/W > 2 (where L and W are the grain length and width, respectively) were classified as columnar, following the criterion proposed by [31].

Microhardness measurements were performed on transverse sections using a Buehler microindenter with a 500 g load and 10 s dwell time. Electrochemical corrosion tests were performed using a conventional three-electrode cell configuration, where the deposited bead acted as the working electrode, graphite as the counter electrode, and Ag/AgCl as the reference electrode. Measurements were carried out using an AUTOLAB CT potentiostat in a 3.5 wt.% NaCl solution at room temperature, in accordance with ASTM G3. Prior to testing, the open circuit potential (OCP) was monitored for 1 h to ensure electrochemical stabilization. Electrochemical impedance spectroscopy (EIS) measurements were then conducted over a frequency range of 20 kHz to 100 mHz. Subsequently, potentiodynamic polarization (PDP) tests were performed by scanning the potential from − 0.25 V to + 0.25 V relative to the stabilized OCP at a scan rate of 0.001 V/s.

3 Results

3.1 Effect of process parameters on bead geometry

Figure 2b shows the nine-deposited single beads. Clear penetration of the AISI 308LSi deposit into the substrate can be observed, appearing more pronounced than reported in previous studies by Karamimoghadam et al. [17] for 308L stainless steel. Figure 3 shows the main effects plots derived from Table 4, illustrating the influence of processing parameters on the responses output: a) melt-in depth (D), b) contact angle (CA), c) bead height (BH) and d) bead width (BW). The results indicate that current is the primary factor controlling the melt-in depth, showing a monotonic and nearly linear increase in penetration with increasing current. In contrast, TS exhibits an inverse relationship with penetration, as higher TS reduce the Heat Input per unit length and consequently limit the melt-in depth. GFR has a relatively minor influence on penetration, although the maximum melt-in depth is observed at the intermediate level of 17.5 L/min. regarding the contact angle, TS appears to be the dominant parameter, with the results showing a quasi-linear increase in contact angle as TS increases. Current and GFR exhibit only a slight increasing effect on the contact angle, indicating a secondary influence. In terms of bead geometry, TS is also the governing parameter controlling both bead height and bead width. Both dimensions decrease in a quasi-linear manner with increasing TS. Current shows a secondary effect, leading to a progressive increase in both bead height and bead width as the current increases. GFR presents an optimal condition at 17.5 L/min, where the maximum bead dimensions are obtained, while higher flow rates slightly reduce these dimensions.

3.2 Grey relational analysis (GRA)

For optimal deposition conditions, the following responses were considered: heat input (HI), where lower values minimize heat accumulation. Bead height (BH), where higher values favor efficient wall building with reduced material consumption. Bead width (BW), where lower values promote thin-wall fabrication. And contact angle (CA), where values greater than 115° ensure stable bead geometry [32]. To evaluate the combined effect of these responses, GRA was applied following the procedure described by Goka et al. [30]. The experimental responses were first normalized within the range [0–1], and the appropriate larger-the-better or smaller-the-better criteria were applied. The Grey relational coefficient and the Grey Relational Grade (GRG) were then calculated to rank the experimental runs. The results in Table 4 show that GRG values closer to one correspond to optimal conditions. The analysis indicates that the most favorable processing parameters for AISI 308LSi deposition are 130 A, 0.48 m/min, and 20 L/min, corresponding to sample E3 (179 J/mm), which exhibits the highest GRG. This condition is associated with the lowest HI and the highest contact angle 131°, leading to improved bead geometry. In contrast, sample E4 (377 J/mm) present the last rank. These results suggest that although HI strongly influences bead formation, it is not the only factor governing bead geometry. To validate the optimized parameters, additional metallurgical, mechanical, and corrosion characterizations were conducted.

3.3 X-ray diffraction analysis

Figure 4 shows the X-Ray diffraction (XRD) patterns of the studied samples. The diffractograms reveal the presence of characteristic peaks corresponding to the face-centered cubic (FCC) γ-austenite and body-centered cubic (BCC) δ-ferrite phases, which are typical of austenitic stainless steels. The γ-austenite phase was indexed at 2θ angles of 51.56° (111), 60.05° (200), 89.90° (220), 111.64° (311), and 119.49° (222), while the δ-ferrite phase was identified by the peak at 52.67° (110). The as-received wire exhibited a strong preferred crystallographic orientation along the (220) γ plane [23]. In the deposited samples, two distinct grain orientation behaviors were observed. Sample E5 (278 J/mm) displayed a pronounced preferred orientation along the (200) γ orientation, whereas the E3 (179 J/mm) remaining samples showed a dominant (111) γ orientation. Such variations in crystallographic texture development during welding or additive manufacturing processes have been previously reported in austenitic stainless steels [24]. These differences are likely associated with variations in thermal gradients, cooling rates, and solidification conditions during deposition.

3.4 Microstructure and microtexture analysis

Figure 5 displays SEM micrographs of samples a) E7, b) E5, and c) E3, along with (d) the EDS point-analysis spots in sample E5. The SEM micrographs reveal a predominant austenitic matrix including residual δ-ferrite that exhibits different morphologies. Dark, nearly round shapes corresponding to porosity are observed in all specimens (E7, E5, and E3). The sizes of the pores vary from around 0.1 to 5 μm.

Table 5 summarizes the EDS-measured elemental composition of sample E5 in weight percent. The detected major alloying elements are consistent with the nominal feedstock composition reported by the manufacturer in Table 1. EDS point analysis indicates compositional differences between the matrix and the residual ferrite. Spot 1, located in the

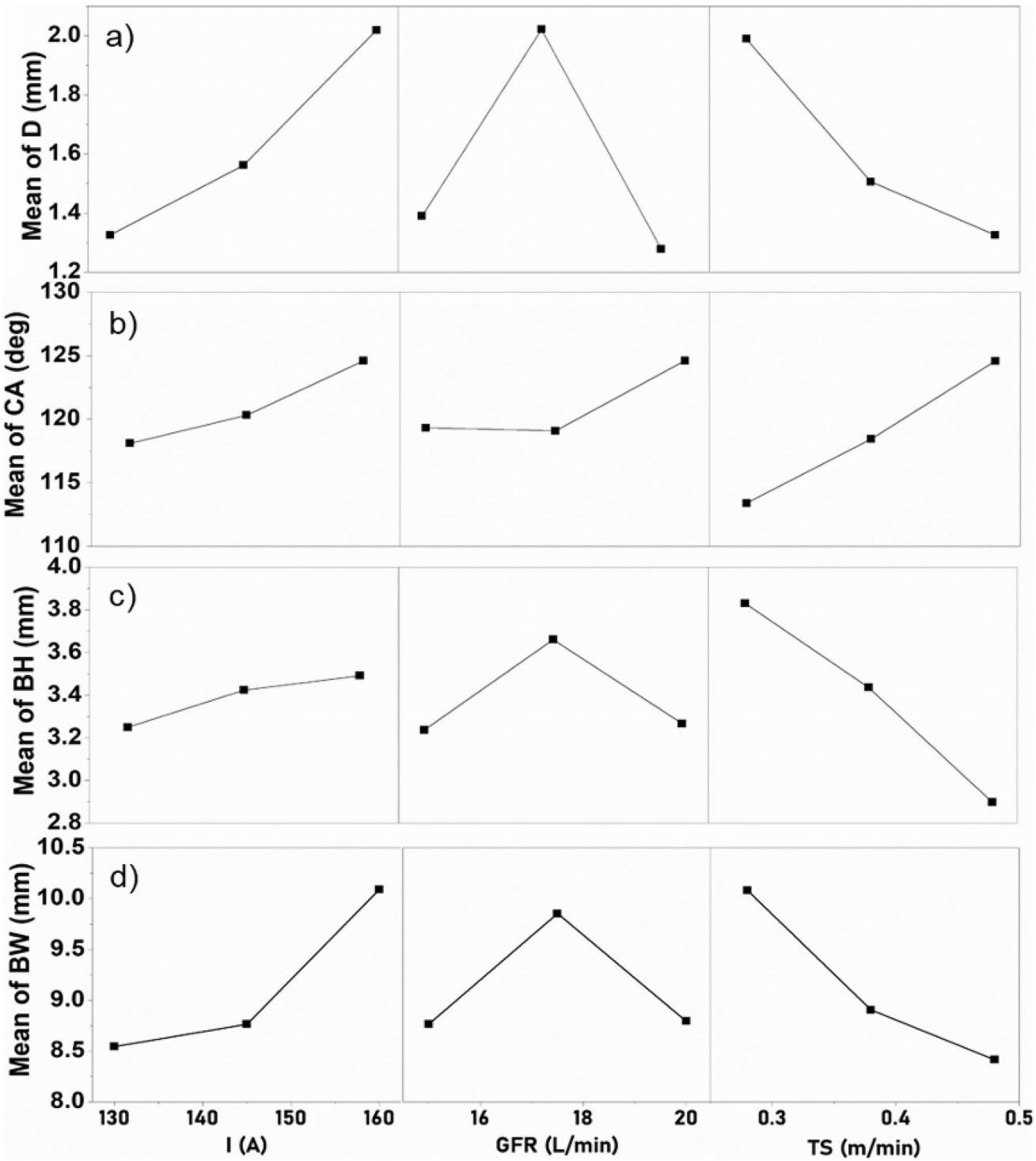


Fig. 3 Main effects of processing parameters(I,TS and GFR) on **a** Melt-in-depth, **b** Contact angle, **c** Bead height and **d** Bead width

austenitic matrix, shows a relatively higher Ni content. Conversely, Spot 2 is characterized by Cr enrichment and a lower Ni concentration, highlighting the distinct chemical compositions of the austenitic matrix and residual δ-ferrite.

Figure 6 shows the locations where EBSD maps were acquired. A grain shape analysis was then performed based on the EBSD data. Figure 7 presents the resulting area fractions of equiaxed and columnar grains for the nine samples (E1 to E9), arranged in ascending order of heat input along with the corresponding experimental conditions. The results show that the surface fraction of equiaxed grains generally varies between 8 to 18%. The lowest fractions are observed

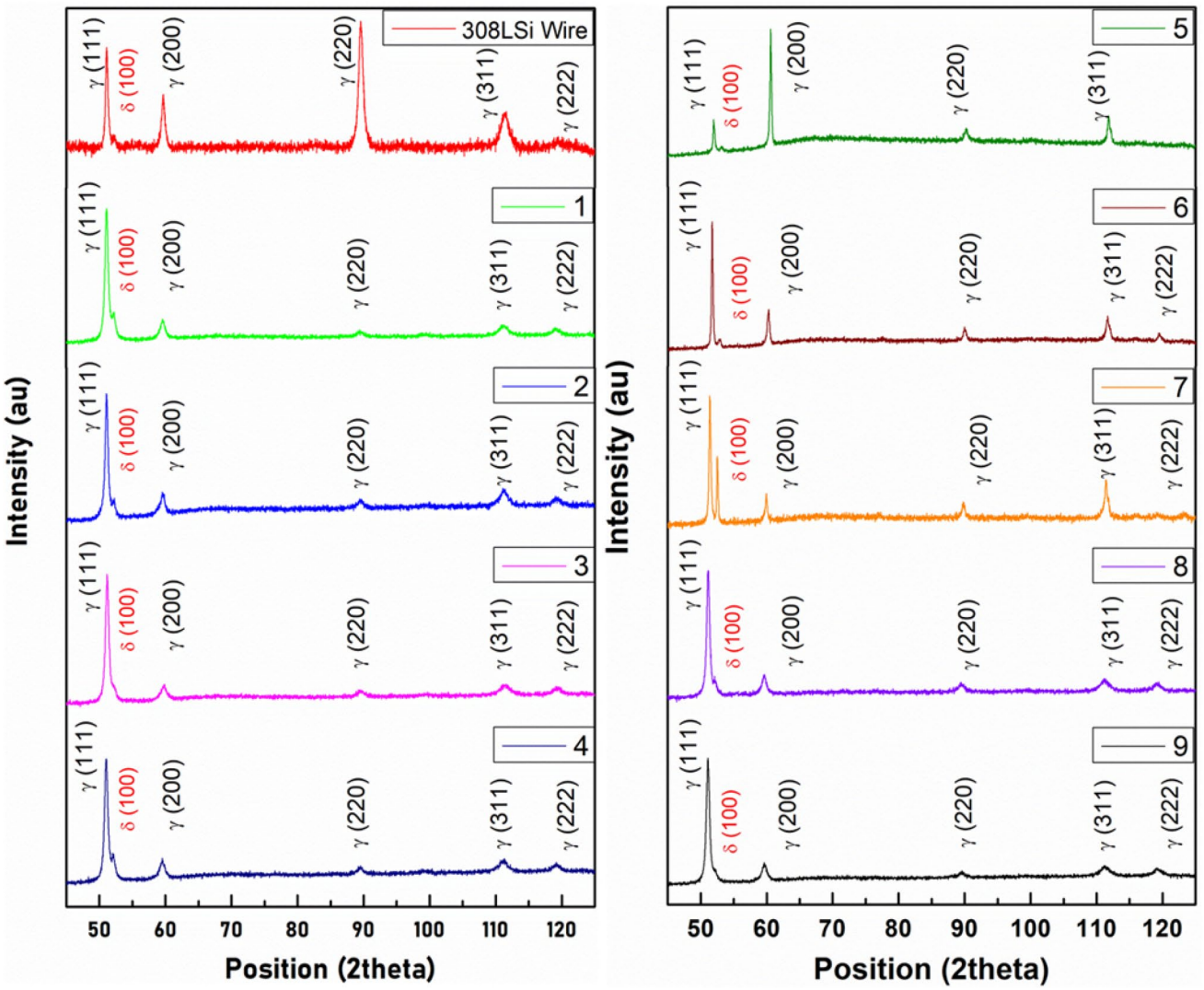


Fig. 4 XRD Spectrums of the as received 308LSi wire and the studied Samples (E1-E9)

for samples E6 (202 J/mm) and E1 (307 J/mm), while the maximum value is obtained for sample E7 (447 J/mm). Furthermore, sample E5, produced with an intermediate HI of 278 J/mm, exhibits an equiaxed grain fraction of 16%, corresponding to the second-highest value. Intermediate values, ranging from 13 to 14%, are also observed for samples E2 (226J/mm), E9 (260J/mm) and E8 (329J/mm), while samples E3 (179 J/mm) and E4 (377J/mm) exhibit equiaxed grain fractions close to 10%.

In the following analysis, we will be focusing on three weld beads: E3 (179 J/mm), E5 (278 J/mm), and E7 (447 J/mm). The XRD results revealed significant differences among these three conditions, while the other beads showed similar diffractograms. These samples correspond to distinct HI levels: E3 (179 J/mm) represents the lowest HI, E5 (278 J/mm) an intermediate HI, and E7 (447 J/mm) the highest HI. Figure 8 presents Inverse pole Fig. (IPF) orientation maps for a)E3, b)E5,c)E7 and d) misorientation angle frequency for (E3,E5 andE7). In all samples, the microstructure is mainly composed of large columnar grains with noticeable branching, typical of directional solidification during the WAAM process. This morphology results from the strong thermal gradients and directional heat flow that develop during solidification of the molten pool. In addition, a fraction of equiaxed grains is observed near the top of the bead. The misorientation distributions reveal a predominance of Σ3 boundaries in samples E3 and E5, with E3 exhibiting the highest fraction. The Σ3 boundaries correspond to a 60° misorientation about the ⟨111⟩ crystallographic axis, which is characteristic of coherent twin boundaries in face-centered cubic (FCC) materials. The measured fractions of Σ3 boundaries are 19% for E3 and 17% for E5. This finding is in good agreement with previous studies on 308L stainless steel, which report

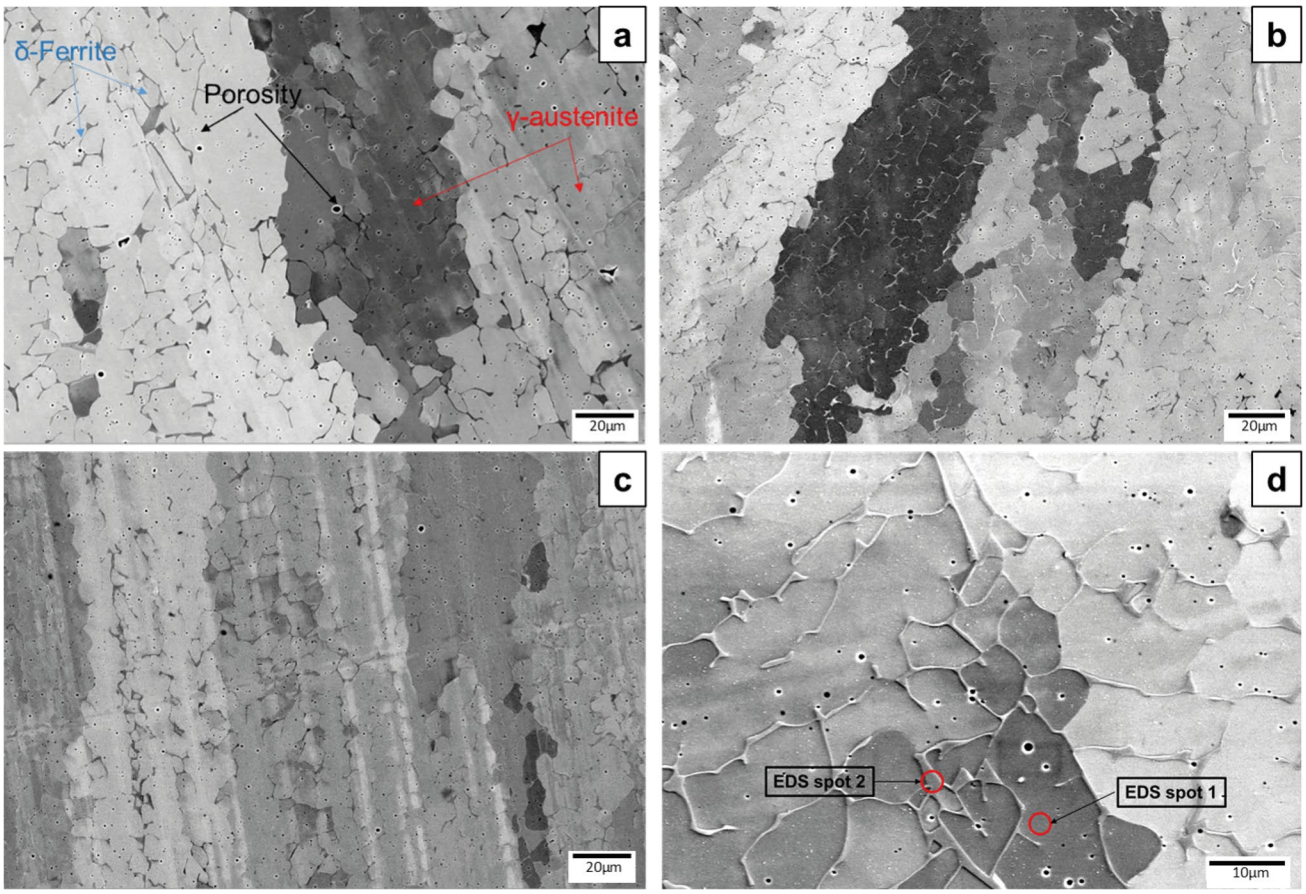


Fig. 5 Micrographs of samples **a** E7 **b** E5 **c** E3 and **d** EDS spots from E5

Table 5 EDS analysis of Sample E5

EDS SPOT 1						
Element	SiK	MoL	CrK	MnK	FeK	NiK
Weight %	0.7	0.21	17.42	1.63	69.72	10.31
EDS SPOT 2						
Element	SiK	MoL	CrK	MnK	FeK	NiK
Weight %	0.71	0.32	23.50	1.37	68.93	5.17

similar trends [33]. In contrast, sample E7, produced under the highest heat input (HI), exhibits a uniform distribution of misorientation angles and a less pronounced Σ3 peak (15%). Figure 9 shows the inverse pole figures (IPF) along the X, Y, and DD directions for the FCC austenite phase in the three samples (E3, E5, and E7). The inverse pole figure analysis reveals differences in texture intensity among the samples. Sample E3, produced with low HI and TS of 0.48 m/min, exhibits the lowest texture intensity (2.6 MRD). Sample E7, produced with high HI and a TS of 0.28 m/min, shows approximately similar texture intensity to E3 but lower than E5 (4.2 MRD). Sample E5, produced with intermediate HI and TS of 0.38 m/min, exhibits the strongest crystallographic texture, characterized by the highest texture intensity among the three conditions.

3.5 Microhardness and corrosion evolution

Microhardness measurements were conducted at four distinct sites along each bead, as seen in Fig. 1a, with two indentations made at every site. The microhardness measurements illustrated in Fig. 10 span from 170 $HV_{0.5}$ to 205 $HV_{0.5}$, aligning with the typical values documented for austenitic stainless steels in a welded state [34]. A local difference in microhardness is detected across each specimen throughout positions P1 to P4, with a significant disparity in average hardness noted between samples, such as E3 and E7. For these specimens, the microhardness demonstrates a general decline as HI increases, consistent with [35].

Figure 11 and Table 6 present the corrosion current I_{cor} and corrosion potential E_{cor} for the deposited samples.

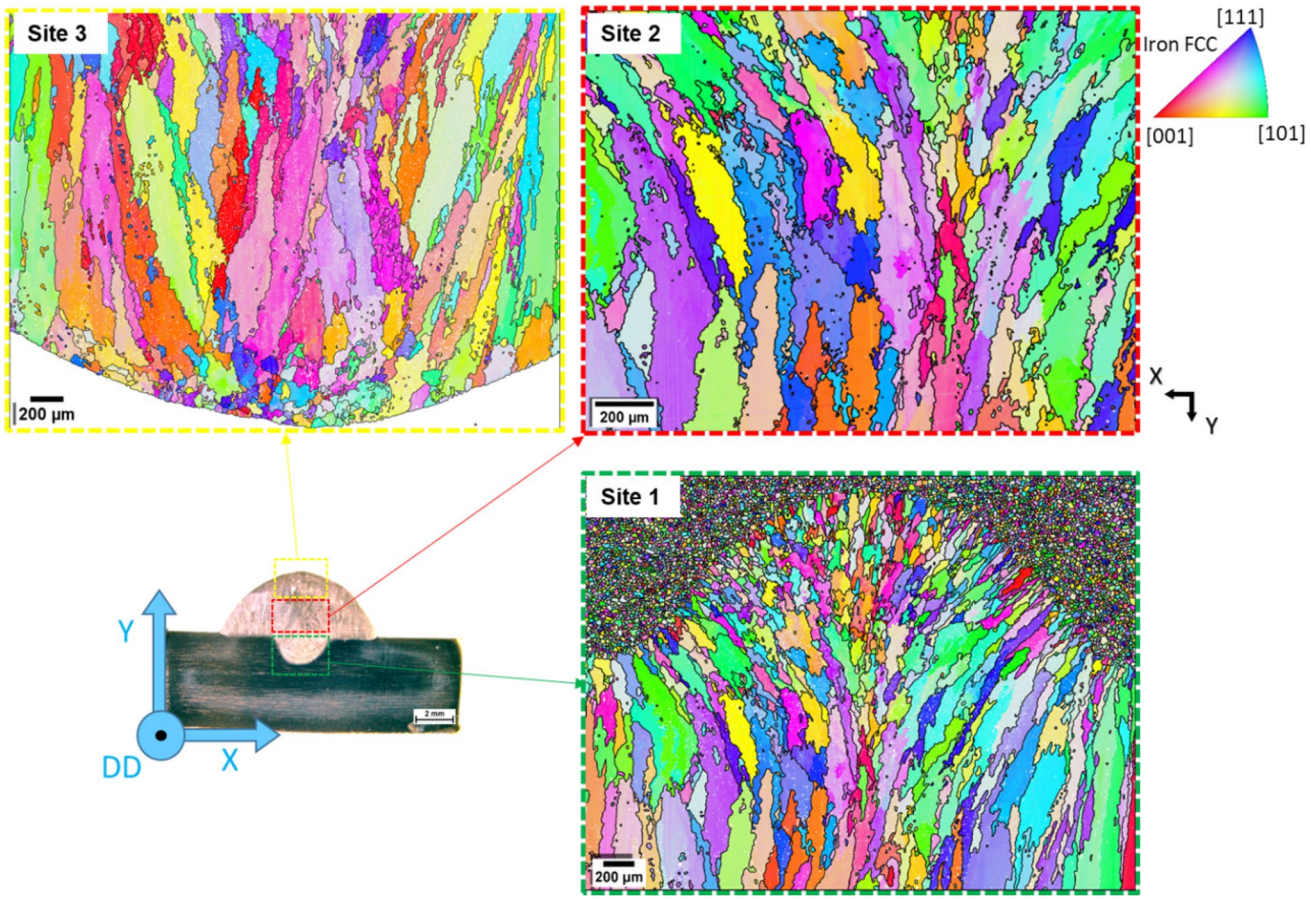


Fig. 6 EBSD acquisition locations and corresponding inverse pole figure (IPF) maps obtained at three representative regions (Sites 1–3) of the WAAM deposited beads

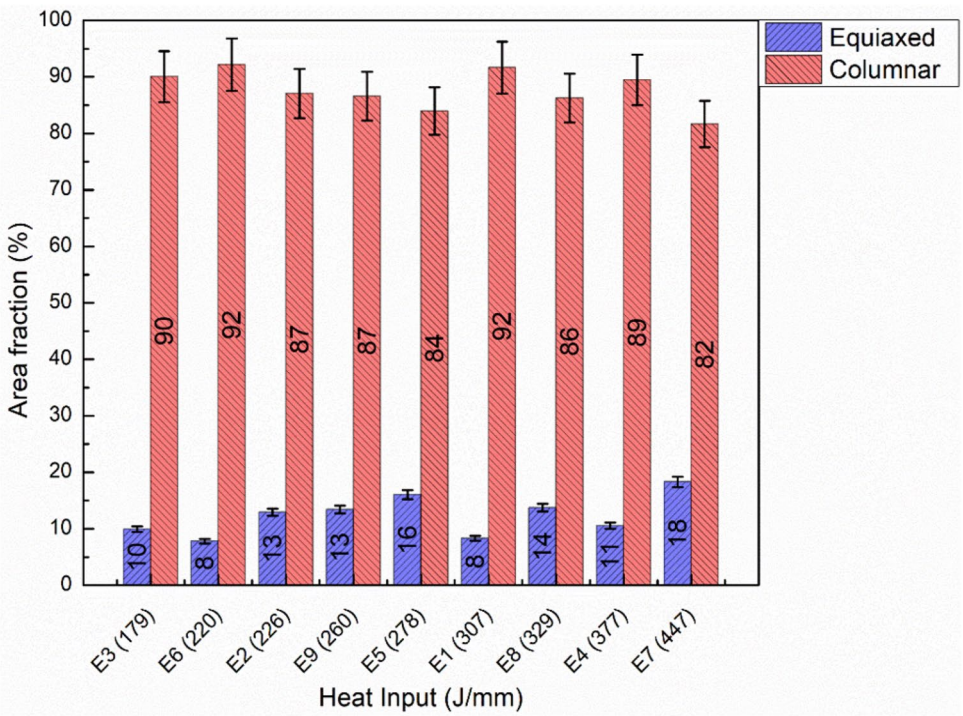


Fig. 7 Grain area fraction distribution as a function of heat input. Samples are classified from the lowest heat input to the highest

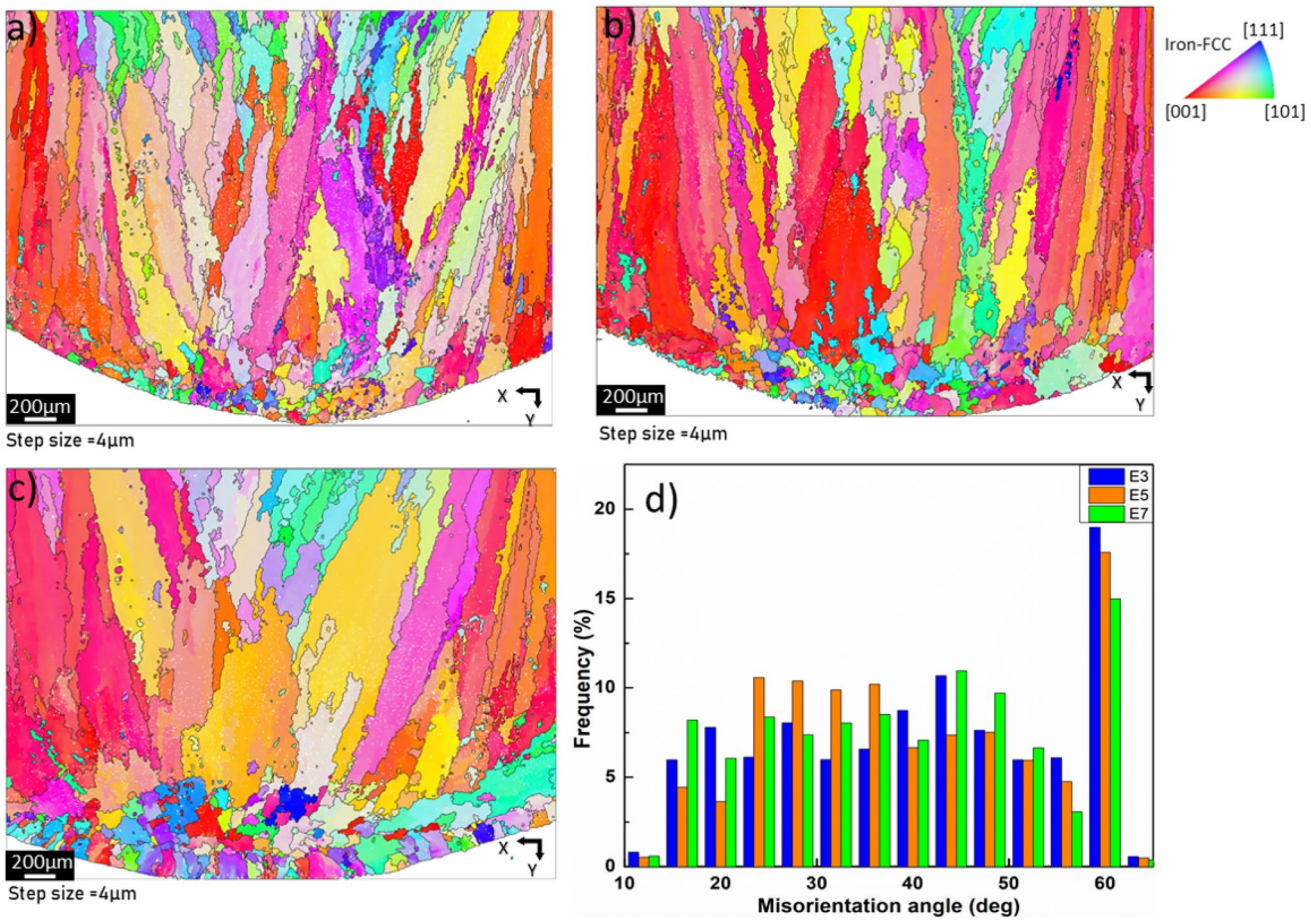


Fig. 8 Inverse pole figure (IPF) orientation maps for **a** E3, **b** E5,**c** E7 and **d** misorientation angle frequency for (E3,E5 andE7)

Sample E5 exhibits the lowest I_{cor} = 1.47E-08 A/cm^2 and the most positive E_{cor} = -0.185 V, indicating the highest corrosion resistance, followed by E3. Sample E7 shows the highest I_{cor} = 1.55E-7 A/cm^2 and the most negative E_{cor} = -0.28V, corresponding to the lowest corrosion resistance While porosity has a major effect on corrosion behavior, the pore fraction in the three samples is reported in Fig. 12.

4 Discussion

As shown in Fig. 2, the penetration depth appears to be more pronounced, reaching a maximum value of 2.31 mm, compared with that reported by Karamimoghadam et al. [17]. The observed behavior can be attributed to the combined effects of thermocapillary fluid flow and the specific composition of the current studied material. The thermocapillary driven fluid flow theory, initially proposed by Heiple et al. [36] or stainless steel welds, establishes that inward fluid flow promotes deeper weld penetration. The presence of silicon in the alloy significantly enhances this effect. As reported in [37], silicon reduces the melt pool viscosity, thereby increasing fluidity. This enhanced fluid flow, combined with localized heat accumulation, enables deeper penetration of the molten pool into underlying layers. Meanwhile, no defect or visible porosity were detected. As can be seen in Fig. 3a, the increase in melt-in depth with elevated current can be attributed to the higher heat input (HI). As indicated by [38] this increase in current generates more thermal energy at the wire-substrate interface, which enhances heat transfer, enlarges the fusion zone, and ultimately facilitates deeper penetration into the base plate. In contrast, the inverse relationship between TS and melt-in depth in Fig. 3a results from the reduced interaction time between the heat source and the substrate at higher speeds. The relatively minor influence of GFR suggests that shielding conditions play only a secondary role in controlling penetration, with optimal conditions occurring at an intermediate flow rate. In Fig. 3b)The increase in contact angle with TS can be explained by the reduction in HI at higher speeds, which limits excessive lateral spreading of the molten pool and promotes a more stable bead profile. According to [39], at low TS, excessive HI reduces melt viscosity and promotes spreading, whereas higher speeds

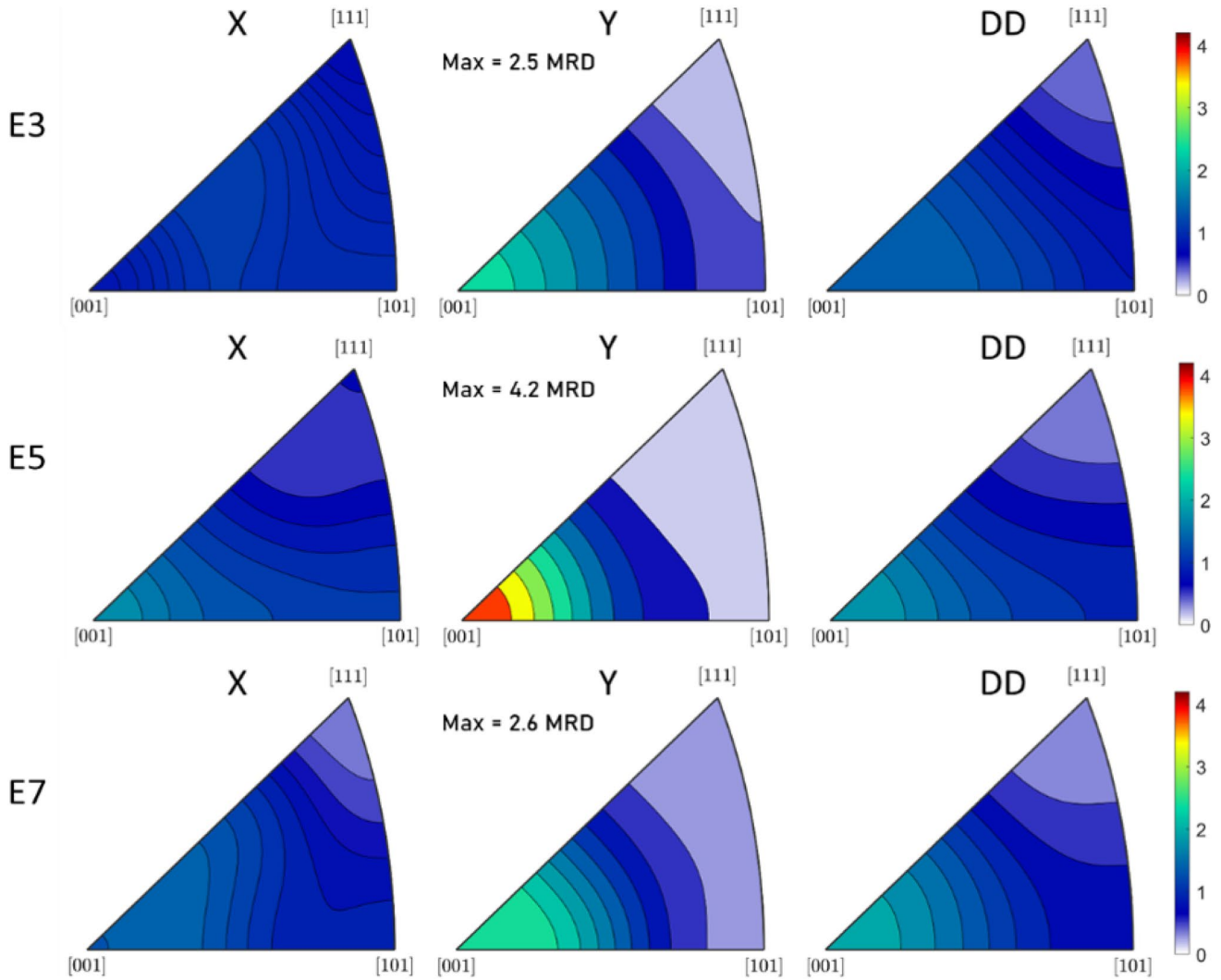


Fig. 9 inverse pole figures (IPF) along the X1, Y1, and Z1 directions for the FCC austenite phase in the three samples (E3, E5, and E7)

stabilize the bead and result in larger contact angles and improved geometric stability. In Fig. 3d, the reduction in bead width and bead height with increasing TS is associated with decreased heat input per unit length and reduced time available for molten pool development. The secondary influence of current, which promotes an increase in bead dimensions, particularly bead width, indicates that higher current mainly enhances lateral spreading rather than vertical material buildup [39]. The existence of an optimal GFR may be related to convective cooling effects or arc stability, which may alter the thermal conditions of the molten pool and consequently affect bead geometry. Overall, although increasing current enhances HI, TS remains the dominant parameter controlling the final WAAM bead geometry.

GRA results enable the identification of optimal process parameters, corresponding to the best trade-off between maximizing bead height (BH) and minimizing bead width (BW). Interestingly, these conditions are also associated with the lowest heat input (HI) and the highest contact angle values, which contribute to maintaining a desirable bead geometry. The highest Grey Relational Grade obtained for sample E3 (179 J/mm) (I = 130A, TS = 0.48m/min and GFR = 20L/min) suggests that this parameter combination provides the most balanced thermal and geometric conditions for AISI 308LSi deposition. These results further indicate that although HI strongly influences bead formation, bead geometry is governed by the combined interaction of multiple process parameters rather than a single factor.

XRD results indicate that the variations in preferred crystallographic orientation observed in the deposited samples are related to differences in the thermal conditions during deposition, as reported by [40]. Where the authors observed that the transition from the (111) texture to the (200) texture is strongly linked to the increase in heat input. Samples E3 (179 J/mm) and E7 (447 J/mm) exhibit very similar diffractograms; however, the δ-ferrite peak intensity is noticeably

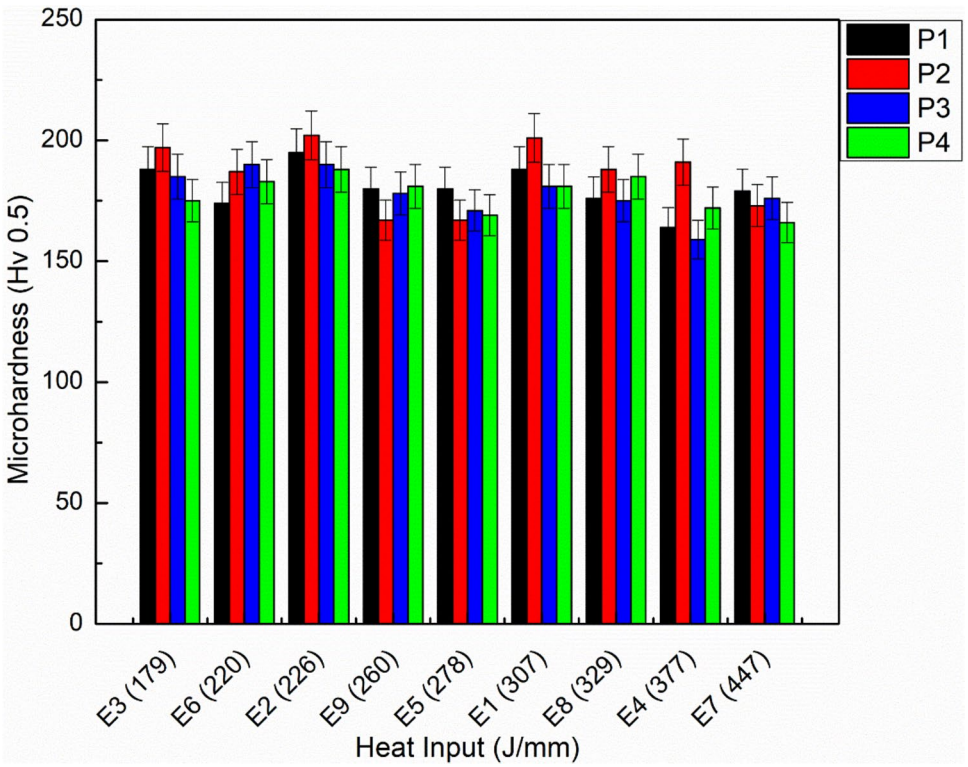


Fig. 10 Microhardness of the nine samples as a function of heat input

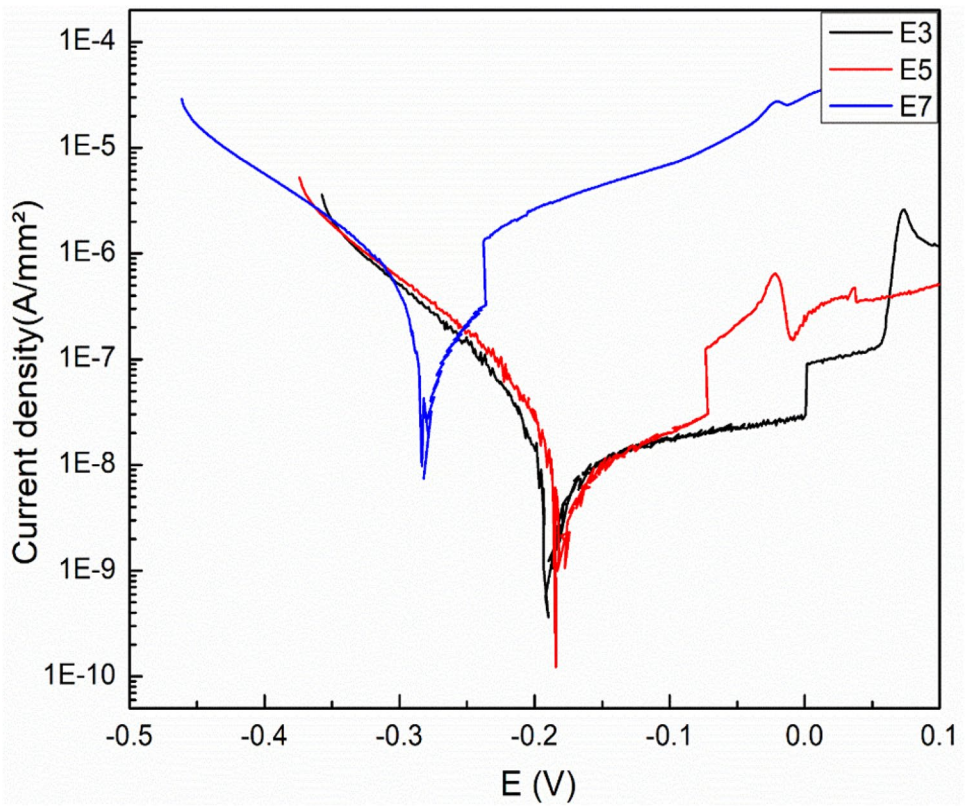


Fig. 11 Current density Vs E of E3, E5 and E7

Table 6 Electrochemical parameters and corrosion rates of E3, E5 and E7 samples

Specimen	E_{corr}(V)	J_{corr} (A/cm^2)	Corrosion rate (mm/year)
E3	-0.1903	3.04×10^{-8}	3.5×10^{-4}
E5	-0.1849	1.47×10^{-8}	1.7×10^{-4}
E7	-0.2828	1.55×10^{-7}	1.8×10^{-3}

higher for E7 (447 J/mm) compared with E3 (179 J/mm) and E5 (278 J/mm). This increased in δ-ferrite peak intensity can be attributed to the lower cooling rate associated with this condition, which may partially suppress the transformation of δ-ferrite into austenite during solidification. Indeed, according to [41] the amount of δ-ferrite that transforms to austenite and the remaining δ-ferrite depends on the cooling rate. According to the Schaeffler diagram for austenitic stainless steels, the solidification mode in this compositional range corresponds to Ferrite–Austenite (FA) solidification mode (Creq/Nieq = 1.62), where δ-ferrite is expected to nucleate first from the liquid phase. During cooling, part of the primary δ-ferrite transforms into austenite, resulting in a predominantly austenitic microstructure containing residual δ-ferrite. As a result, a fraction of the primary δ-ferrite may be retained in the final microstructure [42]. In contrast, sample E5 (278 J/mm) exhibits a distinct preferred crystallographic orientation (i.e. (200)γ texture) compared with the other deposited samples. This condition, corresponding to an intermediate heat input between E3 (179 J/mm) and E7 (447 J/mm), is likely associated with a moderate cooling rate, which may favor the development of a (200)γ texture. This orientation is typically associated with a strong preferential crystallographic texture and has been reported in previous studies on WAAM-fabricated austenitic stainless steels [40].

Based on the analysis of EBSD results in Fig. 7, the grain structure distribution within a weld bead is governed by several processing parameters, including TS, I, and GFR, which together control the solidification conditions in the molten pool [43]. The TS is directly related to the solidification rate R, which can be expressed as $R = Ts\cdot\cos(\alpha)$, where α is the angle between the welding direction and the normal to the solidification front [44] Since α remains constant in the present study, R is proportional to TS. This relationship highlights the influence of process parameters on the growth rate of solidifying grains used in welding and additive manufacturing.

The current I primarily controls the HI (Eq. 1), which in turn affects the thermal gradient G within the molten pool. A higher I generally, results in a higher HI and a higher thermal gradient, thereby significantly modifying the local solidification conditions and the resulting grain morphology. Indeed, Murry [45] shows that higher heat input enlarges the heated volume, prolongs the residence time at high temperature, and alters the distribution of isotherms, thereby changing the local thermal gradients and the shape of the liquidus/solidus fronts that control grain growth directions and morphology in the solidified weld metal. In parallel, the GFR influences heat extraction and cooling conditions at the weld surface through convective heat transfer between the molten pool and the surrounding atmosphere. In parallel, the GFR influences heat extraction and cooling conditions at

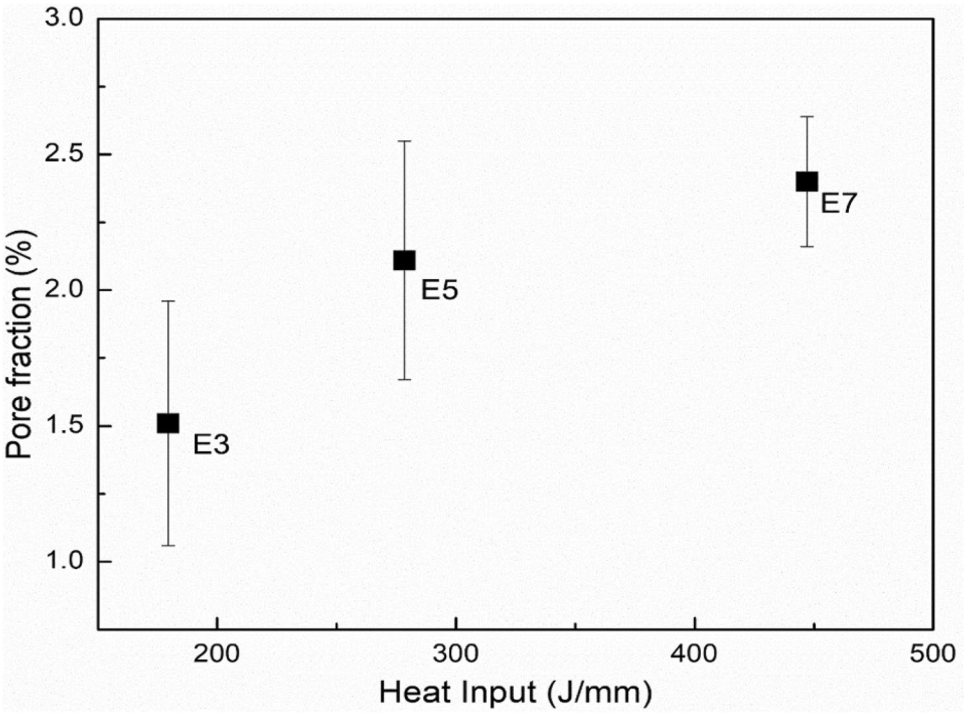


Fig. 12 Porosity as a function of Heat input

the weld surface through convective heat transfer between the molten pool and the surrounding atmosphere. Wells and Lukens [46] showed that increasing convective gas flow by jet impingement significantly modifies weld pool and weld pool surface temperature, cooling rate and thermal gradients near the solid–liquid interface, which in turn controls the solidification macrostructure of the weld metal.

As mentioned earlier, the highest fraction of equiaxed grains for the nine samples is predominantly located near the top surface of the weld bead, where interaction with the shielding gas enhances heat dissipation and increases the local cooling rate. Samples with lower equiaxed grain fractions, such as E1 (307 J/mm) and E6 (220 J/mm), correspond to a lower GFR (15 L/min), which leads to less efficient convective cooling at the surface. In contrast, higher GFR (around 20 L/min) promote an increase in the equiaxed grain fraction by reinforcing surface cooling and promoting constitutional undercooling ahead of the solidification front [46].

Furthermore, the HI significantly influences the balance between the thermal gradient G and the solidification rate R, which is a key factor in the Columnar to Equiaxed Transition (CET) [47]. At higher HI values, the resulting thermal conditions promote strong temperature gradients within the melt pool, favoring columnar growth in the bulk of the weld bead. However, the upper region of the weld bead, being exposed to more intense convective cooling by the shielding gas, experiences a locally increased cooling rate and solidification rate. This localized enhancement of undercooling favors heterogeneous nucleation of new grains and the formation of equiaxed structures near the top surface, as widely reported in welds and laser-based processes [48]. Misorientation analysis reveals that low-HI conditions (E3 = 179 J/mm) in our process lead to an increased fraction of Σ3 twin boundaries, in line with Nie et al. [49] who showed that, under suitable solidification conditions in laser DED stainless steels, a dense network of Σ3 twin boundaries can form directly during solidification. Indeed low HI conditions, which lead to higher cooling rates and stronger thermal gradients, are conductive to solidification twinning in FCC materials with low stacking fault energy [50]. In contrast, the highest HI E7 (447 J/mm) produces a more uniform misorientation distribution with a lower fraction of Σ3 twin boundaries, indicating more random grain boundary structures due to slower cooling and larger melt pools. The evolution of crystallographic texture with HI can be explained by the interplay between thermal gradient (G) and solidification rate (R). Low HI E3 (179 J/mm) and relatively high TS lead to rapid solidification with limited time for competitive grain growth, resulting in weaker texture. High HI E7 (447 J/mm) produces larger melt pools and reduced thermal gradients, promoting orientation dispersion and partially equiaxed grains, leading to moderate texture intensity. The intermediate condition E5 (278 J/mm) provides a favorable G/R regime, balancing thermal gradient and solidification rate, which enables more efficient competitive growth of ⟨100⟩-oriented grains and thus the strongest crystallographic texture observed in this study. These trends are consistent with recent work on WAAM fabricated 316L stainless steel, [51]. Where variations in heat input modify the balance between columnar and equiaxed grains and produce strong ⟨001⟩ textures aligned with the heat-flow direction. Microhardness measurements reveal a clear intra-bead variation in all deposited samples. This variation reflects the microstructural heterogeneity inherent to the WAAM process, in which local differences in grain size, grain morphology, and crystallographic orientation coexist within a single bead, as directly evidenced by the EBSD analysis. Among the investigated samples, E7 (447 J/mm) and E5 (278 J/mm) exhibit a comparatively lower degree of microhardness heterogeneity, although this behavior is governed by different microstructural mechanisms. E7 (447J/mm), produced under the highest heat-input (HI) condition of 447 J/mm, tends to experience lower thermal gradients within the melt pool. This condition promotes more uniform grain coarsening and consequently results in a comparatively homogeneous overall microstructure [52]. In contrast, the relatively homogeneous hardness distribution observed in E5 (278 J/mm) cannot be attributed to its heat input. Rather, it is associated with a strong preferential crystallographic orientation close to (200), corresponding to a dominant ⟨001⟩ grain orientation throughout the bead. This pronounced texture may influence the hardness distribution in two ways: first, by promoting microtextural homogeneity within the bead and second, through the orientation dependence of indentation hardness. Previous studies on directed energy deposited austenitic stainless steels have shown that grains oriented along ⟨001⟩ consistently exhibit the lowest indentation hardness values, whereas ⟨111⟩ oriented grains exhibit the highest values [53]. This can explain the homogeneity of the values of microhardness (P1 = 180 $HV_{0.5}$, P2 = 167 $HV_{0.5}$, P3 = 171 $HV_{0.5}$,and P4 = 169 $HV_{0.5}$) and their lower magnitude compared with those measured in the other deposited beads. Furthermore, EBSD maps reveal the coexistence of equiaxed grains near the top surface of the weld bead and columnar grains in the central region. The highest microhardness values are consistently recorded at position P2 (from 173 $HV_{0.5}$ to 202 $HV_{0.5}$), which corresponds precisely to the zones where equiaxed grains are most frequent. This spatial correlation can be interpreted using a Hall–Petch-type relationship: the finer grain size associated with the equiaxed morphology increases grain-boundary density, thereby restricting dislocation motion and increasing local hardness [54].

The corrosion resistance ranking of the samples follows E5 (278 J/mm) > E3 (179 J/mm) > E7 (447 J/mm), as indicated by both E_{corr} and I_{corr} values reported in Table 6. This ranking is consistent with the corrosion rates, where E3 (3.5×10^{-4} mm/year) exhibits approximately double the corrosion rate of E5 (1.7×10^{-4} mm/year), while E7 (1.8×10^{-3} mm/year) shows a corrosion rate roughly ten times higher than that of E5. Although lower heat inputs are generally reported to improve corrosion resistance in stainless steels, Wang et al. [55] investigated WAAM fabricated 316L specimens and showed that increasing the heat input from 2.41 to 3.59 kJ/cm leads to grain coarsening and a deterioration in corrosion resistance. The highest heat input exhibited the poorest corrosion resistance and the highest corrosion current density. In contrast, the present study shows the best corrosion performance was obtained under the intermediate heat input condition (E5). This result suggests that heat input alone does not control the corrosion response. Instead, the corrosion behavior is more closely associated with the microstructural features developed under each deposition condition. The superior corrosion performance of E5 correlates with its highest fraction of special grain boundaries, including 17% Σ3 boundaries. Grain-boundary-engineering studies on super-austenitic stainless steels [56] have shown that increasing the fraction of Σ3 twin boundaries and breaking the random HAGB network both enhance corrosion resistance and promote the formation of denser, Cr and Mo rich passive films. In this context, the higher density of low-energy Σ3 twin boundaries in E5 is expected to further improve its corrosion resistance by disrupting the continuity of susceptible random boundaries and stabilizing the passive film. Finally, E7, deposited at the highest heat input of 447 J/mm, exhibits the lowest corrosion resistance. This condition not only shows the highest porosity level Fig. 12, which increases the number of potential corrosion initiation sites and the effective surface area exposed to the corrosive environment, but also presents the lowest fraction of special grain boundaries (Σ3 boundaries). The combined presence of higher porosity and a lower fraction of potentially protective Σ3 boundaries may therefore explain the inferior corrosion performance of E7.

5 Conclusion

The present study investigated the influence of CMT-WAAM process parameters on the bead geometry, microstructure, microhardness, and corrosion performance of AISI 308LSi stainless steel. The main conclusions are as follows:

The deposition quality and bead geometry are strongly dependent on process parameters. Silicon promoted bead penetration, while travel speed primarily governed bead morphology and current influenced penetration depth. A shielding gas flow rate of 17.5–20 L/min provided stable deposition conditions and effective surface cooling. Grey relational analysis identified 130 A, 0.48 m/min, and 20 L/min as the optimal parameter combination for the selected responses, providing a balanced compromise between bead geometry and process stability.

The process parameters strongly governed microstructural evolution through their effects on heat input, thermal gradient, and solidification rate. These thermal conditions, in turn, influenced grain growth, twin-boundary formation, and crystallographic texture. Lower heat input promoted grain refinement, a higher fraction of Σ3 twin boundaries, and weaker crystallographic texture, whereas higher heat input favored columnar grain growth and increased microstructural heterogeneity. The intermediate heat input condition E5 (278 J/mm) exhibited a favorable G/R balance, promoting competitive dendritic growth and a pronounced ⟨001⟩ texture. Microhardness was closely associated with grain morphology and crystallographic characteristics. E5 showed a relatively homogeneous hardness distribution, with values of 180, 167, 171, and 169 $HV_{0.5}$ at P1–P4, respectively.

The corrosion behavior was strongly influenced by microstructural characteristics, particularly grain-boundary features. Among the investigated conditions, E5 exhibited the best corrosion resistance, with an Ecorr = -0.185 V, the lowest Jcorr = 1.47 × 10–8 A/cm^2, and a corrosion rate of 1.7×10^{-4} mm/year. The corrosion rates of E3 and E7 were 3.5×10^{-4} and 1.8×10^{-3} mm/year, respectively, corresponding to approximately 2.1 and 10.6 times higher values than E5. The superior corrosion performance of E5 was associated with its optimized grain-boundary network and higher Σ3 twin boundary fraction, which may have contributed to improved passive film stability.

This work demonstrate that the rational selection of CMT-WAAM processing parameters provides an effective approach for tailoring the process, microstructure, properties relationship of AISI 308LSi stainless steel.

Author contributions Conceptualization: Abdelbasset Berriah, Brahim Mehdi, Riad Badji. Methodology: Abdelbasset Berriah, Brahim Mehdi, Salima Bouvier, Riad Badji. Investigation: Abdelbasset Berriah. Formal analysis: Abdelbasset Berriah, Salima Bouvier. Writing original draft: Abdelbasset Berriah. Writing-review and editing: Brahim Mehdi, Riad Badji, Salima Bouvier, Malek Habak. Supervision: Brahim Mehdi, Riad Badji, Salima Bouvier. All authors approved the final manuscript.

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Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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